EXAM SHIELD

Computer Networks Question Bank

Layers, addressing, routing and core Internet protocols.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 22 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Computer Networks

Layers, addressing, routing and core Internet protocols.

1. Which OSI layer is responsible for routing packets between networks?

A. Data link
B. Session
C. Transport
D. Network

Answer: D

2. Which protocol finds the MAC address that belongs to an IPv4 address on a local network?

A. ICMP
B. DNS
C. ARP
D. NAT

Answer: C

3. How many usable host addresses does a /24 IPv4 subnet provide?

A. 254
B. 255
C. 256
D. 252

Answer: A

4. Which transport protocol provides reliable, ordered delivery of data?

A. UDP
B. IP
C. ICMP
D. TCP

Answer: D

5. Which protocol transfers email between mail servers?

A. SMTP
B. POP3
C. SNMP
D. IMAP

Answer: A

6. Which of these is a link-state routing protocol?

A. RIPng
B. RIP
C. OSPF
D. BGP

Answer: C

7. Which device forwards frames inside a LAN using MAC addresses?

A. Router
B. Repeater
C. Hub
D. Switch

Answer: D

8. What does NAT mainly allow?

A. Encryption of all traffic
B. Automatic IP address assignment
C. Translation of names into addresses
D. Many private hosts to share one public IP address

Answer: D

9. The ping utility sends messages using the _____ protocol. [1 mark]

Answer: ICMP

10. DNS queries normally use port _____. [1 mark]

Answer: 53

11. An IPv6 address is _____ bits long, compared with _____ bits for IPv4. [2 marks]

Answer: 128 | 32

12. Explain the TCP three-way handshake and why a connection needs it. [5 marks]

13. Compare IPv4 and IPv6 in terms of address size, header design and the need for NAT. [5 marks]